

A zero-bandwidth counterexample to Conjecture 00000007158

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Abstract

A real 2×2 matrix has complex spectrum $\{0\}$ while its antisymmetric part has Euclidean operator norm 1. This disproves the conjecture's asserted equality between actual spectral imaginary bandwidth and that norm. The discrepancy persists under the usual positive-factor conventions for bandwidth and antisymmetric part.

1 Statement and interpretation

The English source asserts: “The imaginary-part bandwidth of the spectrum is the norm of the antisymmetric part, and the minimizer of the norm is the normal matrix.” The accompanying Chinese version likewise states the bandwidth–norm equality. The exact bilingual source is included as `conjecture.md`.

For a real square matrix A , take its eigenvalues over $\mathbb{C}$, and put

$$K(A) = \frac{A - A^\top}{2}, \quad W(A) = \max_{\lambda \in \text{Spec}(A)} \text{Im } \lambda - \min_{\lambda \in \text{Spec}(A)} \text{Im } \lambda.$$

The norm $\|K(A)\|_2$ is the operator norm induced by the Euclidean vector norm. Thus the first assertion includes the universal equality $W(A) = \|K(A)\|_2$. No normality or diagonalizability hypothesis on A is stated. The subsequent assertion about a normal minimizer does not supply such a hypothesis.

2 Exact counterexample

Consider

$$N = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \quad K(N) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} =: J.$$

Proposition 1. *The matrix N satisfies $\text{Spec}(N) = \{0\}$, $W(N) = 0$, and $\|K(N)\|_2 = 1$. In particular $W(N) \neq \|K(N)\|_2$.*

Proof. For every $z \in \mathbb{C}$,

$$\det(zI - N) = \det \begin{pmatrix} z & -2 \\ 0 & z \end{pmatrix} = z^2.$$

The matrix $zI - N$ is invertible exactly when $z \neq 0$. Thus its algebraic spectrum, equivalently its set of complex eigenvalues, is exactly $\{0\}$. This excludes every other eigenvalue. Both imaginary extremes are zero, hence $W(N) = 0$.

For every real or complex vector (x, y) , we have $J(x, y) = (y, -x)$ and

$$\|J(x, y)\|_2^2 = |y|^2 + |-x|^2 = |x|^2 + |y|^2 = \|(x, y)\|_2^2.$$

Thus J is an isometry, and its operator norm is exactly 1: it is at most 1 on all vectors and attains 1 on $e_1 = (1, 0)^\top$. Equivalently, $J^*J = I$ and $\|J^*J\|_2 = \|J\|_2^2$ yield the same norm calculation. The asserted equality would give $0 = 1$, a contradiction. $\square$

3 Conventions and logical scope

This is a strict gap, independent of the familiar factor conventions. The maximum absolute imaginary part and half the full imaginary bandwidth are also zero for N . Using $A - A^\top$ instead of $(A - A^\top)/2$ gives $2J$, whose operator norm is 2. More generally, $J \neq 0$, so every genuine norm on the matrix space assigns it a positive value. The failure therefore does not depend on choosing a special norm or on a missing factor of two.

The same real matrix is also a complex matrix; its transpose equals its adjoint. Accordingly its antisymmetric part equals its skew-Hermitian part. For the Hermitian imaginary-part convention $(N - N^*)/(2i)$, multiplication by a complex scalar of modulus one leaves the operator norm equal to 1.

The conclusion concerns the equality for the actual spectrum written in the conjecture. It does not refute an inequality placing the spectrum inside a norm-controlled strip, nor an equality restricted to a class with extra hypotheses. A containing strip may have positive width even when the actual spectrum has zero width.

The minimization clause does not specify an admissible class or fixed constraints. No optimization problem is invented here. A counterexample to the explicit first assertion already disproves the conjunction as written; no claim is made about possible repaired minimization statements.

4 Formal verification

The accompanying Lean project targets Lean 4.19.0 and Mathlib v4.19.0. It uses Mathlib's algebraic `spectrum` and proves its equivalence to the existence of a nonzero complex eigenvector. Bandwidth is defined as `sSup` minus `sInf` of the imaginary spectrum; for the proved singleton these are exactly the maximum and minimum above. The `Matrix.L2OpNorm` scope selects the Euclidean operator norm, with an explicit bridge to the associated continuous linear map.

The main theorem `bandwidthEquality_false` negates the universal equality. A further theorem negates its conjunction with an arbitrary additional proposition, without assigning a meaning to the minimization clause. The source also proves the radius, positive-scaling, unhalved-part, complex-adjoint, and zero-separating-norm statements used above. All computations are exact symbolic proofs in Lean; no numerical eigenvalue computation or external mathematical computation is required.

The submission includes reproducible build and source-replay records, an audit of theorem axiom dependencies, and an independent semantic review. These records describe local verification; maintainer acceptance is a separate decision.